Introduction to Artificial Intelligence

Sample Study Notes for AI Study Assistant

1. What is Artificial Intelligence?

Artificial Intelligence (AI) is a branch of computer science that develops machines and software that can perform tasks which normally require human intelligence. These tasks include learning, problem-solving, understanding language, recognizing images, and making decisions.

2. Examples of AI

Common examples of AI include voice assistants, recommendation systems, chatbots, face recognition, language translation, and navigation applications. For example, a music application may recommend songs based on what a user listens to.

3. Types of Artificial Intelligence

Narrow AI: AI designed to perform a specific task. Examples include voice assistants and spam filters.
General AI: A theoretical type of AI that would be able to understand and perform a wide range of intellectual tasks like a human.
Super AI: A hypothetical form of AI that would exceed human intelligence.

4. Machine Learning

Machine Learning (ML) is a part of AI. It allows computers to learn patterns from data and improve their performance without being explicitly programmed for every situation. For example, an email system can learn to identify spam messages by studying examples of spam and non-spam emails.

5. Deep Learning

Deep Learning is a part of Machine Learning that uses multi-layered artificial neural networks. It is useful for tasks such as image recognition, speech recognition, and language processing.

6. Natural Language Processing

Natural Language Processing (NLP) is a field of AI that helps computers understand, interpret, and generate human language. Chatbots, translation tools, and voice-to-text systems use NLP.

7. Benefits of AI

AI can process large amounts of data quickly, automate repetitive tasks, support decision-making, work continuously, and help improve services in education, healthcare, transport, and business.

8. Limitations and Responsible Use

AI systems can make mistakes, reflect bias in their training data, and produce incorrect information. Users should check important answers, protect private information, and use AI fairly and responsibly. AI should support human judgment rather than replace careful thinking.

9. Quick Revision Questions

1. What is Artificial Intelligence?
2. Give three examples of AI.
3. What is the difference between Narrow AI and General AI?
4. What is Machine Learning?
5. What does NLP stand for?
6. Write two benefits and two limitations of AI.

End of sample notes. Ask your AI Study Assistant questions using this PDF.